

Single-Exponential Pachner Search

and Exact Witness Reuse
for Unknot Recognition

A complete bounded search, a locality theorem,
and independently checked algorithmic improvements

Research report prepared for Vladimir Reshetnikov
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Main result. Bounded-upward geometric reachability can be searched in $2^{O(t+U)} \text{poly}(t + U + B)$ time. When the consumed initial tetrahedra lie in an allowed region of size at most r with at most c connected components, the bound becomes $t^c 2^{O(r+U)} \text{poly}(t + U + B)$. These bounds include a polynomial complete endpoint predicate; a more expensive predicate must be charged separately.

Scope. These are proved bounds for a precisely specified family of formal Pachner replacements and transported cocycles. The report does not establish a general quasi-polynomial unknot recognizer. The missing universal locality and coverage theorem is stated explicitly.

Source revision.

66098968e88bba797143ac1bf7ad0ac4c5f697df

Three incoming research archives are reviewed and preserved as dependencies.

Abstract

We replace unrestricted enumeration of local-move orderings by a complete search based on the eventual use of each tetrahedron. Every tetrahedron incarnation has eleven possible commitments: remain unused, be consumed at one of six edges by a 3–2 move, or be consumed at one of four faces by a 2–3 move. Consistently committed enabled moves have disjoint supports and commute. Any trace with at most U upward moves is equivalent to a deterministic committed execution; at most $3t + 5U$ incarnations are involved. Consequently bounded-upward marked reachability admits a single-exponential search. An exact sleep-set implementation inherits this bound while avoiding the large overhead of enumerating inert commitments. A reverse-search generator for regions in the degree-four initial dual graph sharpens the bound to $t^c 2^{O(r+U)}$ when an allowed region has at most r tetrahedra and c components. This gives a constructive quasi-polynomial search class for $c = O(\log t)$ and $r + U = O(\log^2 t)$.

Separately, exact positive witnesses screen redundant deletion queries in normal-surface LP propagation. Minimal support antichains preserve every completed certificate of the deterministic baseline. An explicit algebraic family reduces LP calls from $3s^2 + 2s + 1$ to $2s + 1$ and exact simplex pivots from $6s^3 + 3s^2 + s$ to $3s^2 + s$. The artifact contains additive code, independent replay, genuine triangulation controls, paired measurements, and a diagram-bound positive-certificate adapter. Exhaustion of a bounded coherent family is never reported as knottedness. The results improve concrete search and propagation operations, and identify the geometric statement still needed for a general quasi-polynomial recognition guarantee.

Contents

1	What this continuation establishes	1
1.1	Pinned repository and the incoming work	1
1.2	Relation to the published and announced algorithms	2
1.3	What counts as evidence	2
2	The geometric search model	3
2.1	Face-paired triangulations and legal replacements	3
2.2	Cocycles and the endpoint surface	3
2.3	Endpoints, traces and decision predicates	4
3	Commitments and the single-exponential bound	4
3.1	The relative-boundary commutation lemma	4
3.2	A constant alphabet on each incarnation	5
3.3	Sleep sets give an efficient realization of the same bound	7
3.4	A genuine factorial-to-subset separation	7
4	Locality replaces the full tetrahedron count	8
4.1	A supplied region	8
4.2	Counting and enumerating regions	9
4.3	The constructive locality theorem	10
4.4	A precise remaining global obligation	10
5	Bit complexity and the verification boundary	11
5.1	Binary cocycle values stay controlled	11
5.2	Move and identity costs	11
5.3	Independent endpoint replay	12
5.4	Disc components and diagram provenance	12
5.5	Resource limits and observational guarantees	13
6	Exact witness screening for normal-surface propagation	13
6.1	The query and its exact evidence	13

6.2	Preserving the exact baseline policy	14
6.3	A rigorous quadratic-to-linear LP-call separation	15
7	Implementation and integration	16
7.1	An additive research path	16
7.2	Exact geometric event representation	17
7.3	The three search methods	17
7.4	Representative usage	18
7.5	Policy compatibility and maintained costs	19
8	Reproducible experiments and negative controls	19
8.1	Protocol and interpretation	19
8.2	Finite completeness and independently checked geometry	20
8.3	Independent collapse regions: a geometric control	20
8.4	Exact LP reuse on genuine triangulations	21
8.5	Figure-eight controls expose the unsolved cost	22
8.6	The algebraic separation, measured and proved	23
8.7	What the evidence supports	23
9	A concrete route toward quasi-polynomial recognition	24
9.1	The sufficient coverage theorem we would need	24
9.2	Priority 1: geometric locality and an honest obstruction search	24
9.3	Priority 2: stronger complete geometric search	25
9.4	Priority 3: exact LP discovery and normal-surface structure	26
9.5	Priority 4: evaluation that can disprove the proposed story	27
10	Reproduction, provenance and integration	28
10.1	Archive organization	28
10.2	Run the new focused tests	28
10.3	Replay saved evidence before rerunning timings	29
10.4	The complete regression gate	30
10.5	Apply the proposed additions	30
10.6	Rebuild and verify the report	31
A	Exact pivot counts for the algebraic screening family	31
A.1	Ordered chain systems and the exact initial tableau	31
A.2	The actual block ordering and previous propagations	34
A.3	Summing the exact costs	34
B	Claim and assumption ledger	35
B.1	Useful counterexamples to tempting shortcuts	36
B.2	Research status	36

1 What this continuation establishes

The principal improvement concerns discovery. Given a triangulation, a fixed integral cocycle, and a bound on the total number of expanding Pachner moves, the new search covers every permitted finite trace up to explicit marked combinatorial isomorphism. Its analysis removes an unnecessary $\log t$ factor from the exponent of naive move-string enumeration. More significantly, it allows a locality parameter to replace the full initial tetrahedron count in the exponential factor. The normal-surface LP improvement is independent and can be enabled without changing the mathematical branch policy.

Result	Proved guarantee	Remaining restriction
Commitment encoding	At most 11^{3t+5U} encodings of bounded-upward trace classes	U counts total upward moves
Sleep-set exploration	One explored representative per commuting trace class	Precise local action identities are required
Region search	$t^c 2^{O(r+U)} \text{poly}(t + U + B)$ with a polynomial complete endpoint predicate	A small region must contain the successful initial footprint
LP witness screening	Same completed baseline certificate; fewer deletion LPs	Node LPs and the search tree can remain expensive
Source-bound adapter	Every UNKNOT has independent geometric replay	A failed bounded search is inconclusive

These statements concern operations that are implemented in the accompanying artifact. Their proofs do not assume that an observed success rate extends to all unknots. Nor does the reported single-exponential bound improve the best general exponential classification of unknot recognition. It improves a particular geometric search mechanism and supplies a sharper constructive condition under which that mechanism is quasi-polynomial.

1.1 Pinned repository and the incoming work

The source audit is pinned to the GitHub main revision

66098968e88bba797143ac1bf7ad0ac4c5f697df.

The relevant maintained directory is `Topology/UnknotRecognition`; the three incoming archives are in `docs/incoming`. Their Git object identities and SHA-256 digests are retained in the package provenance. Source files were obtained from this immutable revision, not from a moving working-tree comparison.

The incoming dual-certificate report supplies exact positive-Euler LP propagation, dual replay, and primitive-ray certificates. Its search still restarts a rational LP for every unskipped deletion test. The geometric-transport report supplies formal Pachner moves, signed cocycle transport, local scores, and a diagram-bound disc verifier, but implements a single deterministic downward epoch. The support-sensitive report supplies more economical component and rank-one certificates. Its contributions are retained for reproduction but are not presented as new results here.

Incoming archive	Dependency relevant to this continuation
<code>unknot_dual_certificates_20261009.zip</code>	Exact kernel LP, standard-coordinate propagation and independent proof checker
<code>unknot_geometric_transport_20261009.zip</code>	Cochain transport, geometric score controls and diagram-bound terminal verification
<code>unknot_support_sensitive_certificates_20261009.zip</code>	Support and component certificate infrastructure; reviewed separately from the new claims

The public repository’s current source and these incoming artifacts are unpublished project material [12, 9, 10, 11]. The additive overlay avoids rewriting their existing production functions. The supplied integration script accepts an already present byte-identical dependency and refuses a conflicting file, so a later integration can be reviewed against the pinned baseline.

1.2 Relation to the published and announced algorithms

Lackenby’s 109-page 2021 talk announces an $n^{O(\log n)}$ unknot-recognition algorithm and describes multi-surfaces and Cheeger-region reductions [7]. The July 2026 paper develops a hierarchy algorithm for incompressibility. Its Proposition 9.1 bounds hierarchy-loop work by $L(g + 1)^L$; the refined length bound is linear in a decomposition complexity. The paper does not provide the complete quasi-polynomial implementation sought here [8]. A separate compressed Oxford handout uses a different displayed exponent, so the artifact identifies the exact talk version it cites.

Pachner-graph search and partial commutation of moves have substantial prior work. Burton studies simplification paths and explicitly rearranges certain consecutive moves, including overlapping composite cases [2]. Burton and He prove an existence theorem for unimodal elementary-move sequences [4]. Our proof uses a smaller independence relation—disjoint consumed tetrahedra—and obtains an explicit bound in the total upward count and initial region. No claim is made to have introduced commutation to triangulation algorithms or to dominate every existing Pachner-graph implementation.

Sleep sets are a standard partial-order verification technique [6]. The new argument specializes their required independence conditions to the repository’s exact formal replacements and combines them with the incarnation count. The LP baseline belongs to the normal-surface branching tradition developed, for example, by Burton and Ozlen [5]. The abstract hardness result of Burton and He cautions against inferring an easy general normal-surface search from its linear relaxation [3].

1.3 What counts as evidence

There are three separate standards throughout this report. A mathematical theorem has an explicit model and proof. A correctness claim about the code has independent replay or comparison with an independent finite oracle. A performance claim has a declared workload, caps, raw timings and a paired comparison. These standards answer different questions. A short execution on the coherent obstruction does not prove a universal small-footprint theorem; a factorial operation-count separation does not guarantee that validation overhead is negligible; a completed LP proof does not authenticate a knot diagram unless source geometry is also included.

2 The geometric search model

2.1 Face-paired triangulations and legal replacements

A triangulation is a finite list of abstract tetrahedra with local vertices 0, 1, 2, 3. Each triangular face is either a boundary face or is paired to another face by a full vertex permutation. Global vertices and edges may have multiple local representatives. The existing validator checks the manifold and matching requirements of the normal-surface interfaces. This is the usual generalized triangulation setting used in computational three-manifold topology, rather than a requirement that every global vertex set specify a unique simplex [2].

A legal 2–3 replacement consumes two distinct tetrahedra sharing a face. Their formal union is a triangular bipyramid. It is replaced by the three tetrahedra around the other diagonal, preserving all six labelled boundary facets and their attaching permutations. A legal 3–2 replacement consumes three distinct tetrahedra constituting the entire degree-three star of an interior edge, with the formal triangular-link consistency checked by the existing producer. It performs the inverse relative-boundary replacement.

These conventions are substantive. All independence statements below use the repository’s formal-bipyramid legality. If one additionally imposed strict simplicial-complex conditions forbidding a new edge or face already present elsewhere, disjoint consumed supports would require a fresh independence analysis. The source code does not silently switch between these conventions.

Definition 2.1 (Incarnation and event). A tetrahedron incarnation is one specific tetrahedron from its creation until its destruction. An event is a legal replacement together with the identities of all consumed incarnations and the local edge or face selected in each. Surviving incarnations keep their local vertex labels; replacement outputs receive new identities and the canonical local labels of the producer.

An incarnation is not a row number. After deleting three rows, a survivor’s row number can change even though its identity, local corners and commitment remain unchanged. The implementation keeps this distinction explicit.

2.2 Cocycles and the endpoint surface

The input marking is a list of four integral heights $h_{t,0}, \dots, h_{t,3}$ per tetrahedron. The signed differences on identified edges must agree; they therefore define an integral cocycle. Adding a constant to one height row does not change the marking. The implementation normalizes each row by subtracting its first entry.

For bit-complexity statements, B bounds the bit lengths of both the supplied height entries and their edge differences. If rows are already normalized, the edge data alone suffice. Including the supplied entries charges the input-reading and normalization cost even when a row has an unnecessarily large additive constant.

If a, b, c, d order the corners so that $h_a \leq h_b \leq h_c \leq h_d$, the corresponding coherent normal vector has triangle coordinates $h_b - h_a$ at a and $h_d - h_c$ at d , and the single quadrilateral coordinate $h_c - h_b$ separating $\{a, b\}$ from $\{c, d\}$. All other local coordinates vanish. Ties cause zero coordinates and cause no ambiguity in the resulting normal vector.

The five formal bipyramid heights determine the transported cocycle on the new tetrahedra. This preserves the global cohomology class. It can change the topology of the coherent normal surface:

the incoming transport package gives a checked example in which a disc becomes a disc and two spheres. Thus raw Euler characteristic is not used as a terminal unknot criterion.

Definition 2.2 (Bounded-upward family). For a source (T, h) and integer $U \geq 0$, let $\mathcal{R}_U(T, h)$ be the marked endpoints of all finite legal 2–3/3–2 traces containing at most U total 2–3 events, with the prescribed cocycle transport at every event. The empty trace and all intermediate endpoints are included.

If a trace contains u upward and d downward events, its tetrahedron count is

$$t' = t + u - d. \quad (1)$$

In particular, $d \leq t + u$ and the trace length is at most $t + 2U$. The maximum tetrahedron count is at most $t + U$. These coarse inequalities also cover degenerate stopping points without requiring a special minimum-size constant.

The parameter U is different from the maximum excess tetrahedron count and from the number of upward bursts. Repeated expansion–collapse pairs can have excess height one and arbitrarily many upward events. Our completeness proof permutes independent upward and downward events; it preserves total counts, but does not preserve every prescribed burst pattern.

Symbol	Meaning
n	Number of crossings when a knot diagram is the original input
t	Number of tetrahedra in the source of one geometric search
U	Allowed total number of 2–3 events
F, r	Allowed set of original tetrahedra and its size bound
c	Maximum number of connected components of the allowed region
B	Initial bit-length bound including supplied heights and edge values
Q	Number of outer searches in a conditional recognition analysis

2.3 Endpoints, traces and decision predicates

Two adjacent events with disjoint consumed supports will be shown to commute. Trace equivalence is generated by such adjacent interchanges, carrying local labels and the marking through the induced isomorphisms. Endpoint predicates must respect this transported isomorphism. The existence of an essential-disc component does; a predicate inspecting the arbitrary order of rows in a certificate need not.

The code supports an arbitrary callback for experimentation, but `ENDPOINT_SELECTED` means only that this callback selected a supplied state. A `DISC_FOUND` result additionally carries an independently checked component-disc proof. Source geometry is then needed to turn the latter into `UNKNOT` for a given diagram.

3 Commitments and the single-exponential bound

3.1 The relative-boundary commutation lemma

Lemma 3.1 (Commuting disjoint supports). *Let a, b be simultaneously legal formal replacements in a triangulation, and suppose that their sets of consumed tetrahedron incarnations are disjoint.*

After applying a , the transported event b remains legal; the symmetric statement holds. The results of ab and ba have a canonical combinatorial isomorphism fixing every surviving old tetrahedron and respecting the two replacement regions, their boundary attachments and their transported cocycles.

Proof. Regard each consumed region as a separate formal bipyramid before the global face identifications. Both fillings have the same labelled six-facet boundary. One can replace both fillings simultaneously and then impose the original attaching maps. Replacing them in either order gives exactly this quotient. This also covers the case that a boundary face of one region is attached to a boundary face of the other: the common attaching permutation is retained in the simultaneous construction.

The internal face of a 2–3 event belongs to its two consumed tetrahedra. The complete central-edge star of a 3–2 event belongs to its three consumed tetrahedra. A disjoint replacement changes neither of these internal configurations. Its updates to exterior attachments merely apply the same relative-boundary identifications. Hence the second event stays legal.

For the marking, the five heights in each formal region are reconstructed from the same old edge differences. Changes of additive row constants have no effect. The regions use disjoint old tetrahedron interiors and retain the same boundary edge values. The two independently obtained sets of new rows therefore agree after the stated isomorphism and normalization. No claim that the intermediate coherent surfaces have identical topology is required. $\square$

Disjoint support is sufficient, not necessary. Overlapping moves can also have special relations, as in the earlier Pachner-graph literature [2]; the present proof deliberately requires only the uniformly checked relation in Lemma 3.1.

3.2 A constant alphabet on each incarnation

Assign an incarnation one of the following eleven symbols:

$$\mathcal{A} = \{\text{idle}\} \cup \{\text{down at edge } e : 0 \leq e < 6\} \cup \{\text{up at face } f : 0 \leq f < 4\}.$$

The six edges use a fixed ordering of the pairs of local vertices. An enabled downward event is *ready* if all three source incarnations are committed to their local copy of its central edge. An upward event is ready if both source incarnations are committed to the common face. Upward events are disabled when the total upward budget is exhausted.

Lemma 3.2 (Ready events cannot compete). *Two different ready events consume disjoint sets of incarnations.*

Proof. A shared incarnation would have to select the same symbol for both events. The idle symbol selects no event. A selected local face identifies its unique partner and hence its unique 2–3 event. A selected local edge identifies its entire global edge star and hence the unique legal 3–2 event at that edge. Thus the purported events would be equal. $\square$

The committed search uses a fixed deterministic order of ready events. If one exists, it executes the first, preserves commitments on survivors and leaves new incarnations unassigned. If none exists, it assigns the first unassigned live incarnation one of the alphabet symbols, branching on every choice. If none is unassigned, that branch stops. An assignment need not be consistent with any eventual move; inert assignments are allowed and counted.

The implementation evaluates a state predicate at each newly encountered marked endpoint, before deciding whether to continue. This avoids the extra assumption that a useful surface survives every later downward choice.

Theorem 3.3 (Complete commitment encoding). *Let T have t tetrahedra and let $U \geq 0$. Every permitted trace defining an endpoint in $\mathcal{R}_U(T, h)$ is equivalent to an execution of the deterministic committed scheduler. At most $3t + 5U$ incarnation symbols are needed. Consequently the number of commuting trace classes, including classes of all finite prefixes, is at most*

$$11^{3t+5U}. \quad (2)$$

For $U = 0$, the alphabet can be restricted to seven symbols and the corresponding bound is 7^{3t} .

Proof. Fix a target finite trace. For each incarnation appearing in it, record the local port of the event that eventually consumes it; if it survives to the target endpoint, record idle. Follow these assignments in the deterministic scheduler, interpreting them under the transported local labels.

Suppose an event is currently ready. Consider the first event in the remaining target trace that touches one of its source incarnations. Until that first touch, the ready event's common face or full central-edge star is unchanged. The touched incarnation's commitment specifies the local port of its eventual consuming event. This is the ready event's port, so uniqueness at that port implies that the first touching event is exactly the ready event. All earlier target events have disjoint consumed supports. By Lemma 3.1, the ready event can be moved to the front across all of them. This interchange preserves the endpoint, the marking and the total upward count. Its new prefix cannot exceed the total allowed upward count.

If no event is ready, assign the next unassigned incarnation its recorded symbol. Once the participants of the first remaining target event have received their symbols, that event is ready. Thus this process cannot stop before all target events have been performed, apart from harmless reordering. All target survivors are idle, so it stops at the required endpoint after the assignments are completed. This proves completeness and trace equivalence without asserting confluence of uncommitted downward rewriting.

If the execution has u upward and d downward events, the total number of incarnations is

$$N = t + 3u + 2d \leq t + 3u + 2(t + u) = 3t + 5u \leq 3t + 5U.$$

Each is assigned at most once. Encode the assignments in the deterministic order in which the scheduler requests them, and pad a shorter encoding to length $3t + 5U$ by idle symbols. Replaying a word determines one execution. If two trace classes had the same encoding, they would both commute to that same execution and would be equal. Hence there are at most the number of such words. The same construction applied to any target prefix, giving its survivors idle, proves the assertion for all prefix classes. Omitting upward symbols gives the last statement. $\square$

Remark 3.4 (Assignments are not geometric steps). The encoding algorithm is a proof device as well as a reference implementation. Many assignments lead nowhere. Its tree has at most $(11^{N+1} - 1)/10$ assignment vertices, with polynomially many forced-move operations along an assignment path. It can be much slower on a small triangulation than direct trace reduction. The bound does not justify a claim of practical speed by itself.

3.3 Sleep sets give an efficient realization of the same bound

At a search node maintain a set of sleeping event identities. Explore the enabled events in deterministic order. Skip a sleeping event. After exploring an event a , add a to the local sleep set for later sibling branches. For a child reached by b , retain only sleeping events whose consumed support is disjoint from b 's support. Thus an event is remembered across a replacement exactly when the proved commutation rule permits it to move past that replacement.

Theorem 3.5 (Sleep-set coverage and uniqueness). *With exact incarnation identities, one canonical event per eligible edge or face, and the independence relation of Lemma 3.1, sleep-set exploration visits at most one representative of each commuting trace class. Every reachable marked endpoint has a visited representative. Hence the number of visited trace nodes is at most 11^{3t+5U} , or 7^{3t} when $U = 0$.*

Proof. For uniqueness, suppose equivalent traces first diverge at a state where the earlier branch takes a and a later branch takes b . The later branch starts with a sleeping. Legal interchanges preserve exact event identities, so it must eventually perform that same event a . None of its consumed incarnations can have been destroyed in the intervening prefix: an incarnation never reappears after destruction. Every intervening event thus has consumed support disjoint from a . Initially a is enabled, and Lemma 3.1 inductively keeps it enabled across that prefix. The sleep-update rule retains a throughout, so its eventual occurrence is forbidden. This is a contradiction.

For coverage, choose among the finitely many representatives of any target trace class the first one in the deterministic search order. If it were blocked by a sleeping event, the earlier occurrence that put that event to sleep could be commuted across the intervening independent events, giving an earlier representative of the same class. This contradicts the choice. Equivalently, repeatedly move the earliest blocking event left until no blocking event remains; the finite lexicographic order decreases. Thus the chosen representative is visited. Lemma 3.1 supplies the persistence of all transported enabled events used in this argument.

The total-upward restriction is invariant under these interchanges. Protected initial cells, when present, are also invariant. Apply Theorem 3.3 to the resulting representatives. $\square$

This proof uses the standard sleep-set principle [6], but its hypotheses are checked for these geometric events. The implementation does not merge search continuations merely because their raw triangulations agree. Different remaining upward budgets and sleep sets can permit different continuations. A separate exact state table suppresses repeated endpoint queries only; exploration continues from every scheduled state.

Independence here concerns simultaneously enabled events, or legal adjacent interchanges. It is not a static relation inferred only from the consumed sets of arbitrary historical events: a later event can consume an output born from an earlier event even though their consumed sets are disjoint. The persistence and nonresurrection argument above is what justifies the actual sleep update.

3.4 A genuine factorial-to-subset separation

Suppose p independent downward events are enabled, no other downward event is created, and each can be executed at most once. Unreduced all-prefix enumeration visits

$$N_{\text{naive}}(p) = \sum_{j=0}^p \frac{p!}{(p-j)!} = p! \sum_{k=0}^p \frac{1}{k!}. \quad (3)$$

Every subset of events is a distinct prefix endpoint and all orderings of that subset commute. Sleep exploration visits exactly 2^p nodes. Full maximal orders alone are reduced from $p!$ to one.

The artifact realizes this situation by disjoint inverse bipyramids in a layered solid torus and verifies the move inventory and endpoints. It is therefore an actual geometric control. Equation (3) is a proof under its stated event-inventory hypothesis; finite tests of the construction are reported at the tested sizes. We do not extrapolate the measured construction to a universal claim about all solid-torus triangulations.

4 Locality replaces the full tetrahedron count

The uniform commitment bound still has an exponent linear in t , even when only a small part of a triangulation participates in a useful sequence. The next result charges the exponent to the *initial* region that the trace is allowed to consume. Newly created tetrahedra are included automatically in the incarnation accounting.

Definition 4.1 (Allowed initial region). Let G_T be the simple graph on the original tetrahedra, with adjacency when they share a paired face. An allowed initial region is a set $F \subseteq V(G_T)$. A trace is supported in F if no original tetrahedron outside F is consumed. It may consume every subsequently created tetrahedron. For $r, c, U \geq 0$, let $\mathcal{R}_{r,c,U}(T, h)$ be the union of bounded-upward families supported in regions F with $|F| \leq r$ and at most c connected components in $G_T[F]$.

We use effective integer caps $0 \leq r \leq t$ and $0 \leq c \leq r$; larger supplied caps can be reduced to these values without changing the family. The empty region is included, so zero caps still allow the source endpoint to be checked.

The region is a *cover* of the actual initial consumption footprint. Some members of F may survive. An actual consumed set can have several components while lying inside one connected allowed region. This distinction matches the executable search and is useful when a small connecting corridor is included in the supplied region.

4.1 A supplied region

Theorem 4.2 (Region-sensitive commitment bound). *For a supplied region F of size i , the family supported in F with at most U upward events has at most 11^{3i+5U} commuting trace classes. When $U = 0$, the bound is 7^{3i} . Both the commitment search and the sleep search exhaust this family with polynomial overhead per encoding or representative, apart from endpoint-predicate work.*

Proof. Give every original tetrahedron outside F the forced symbol idle. Such tetrahedra are protected forever. After u upward and d downward events, at least $t - i$ original tetrahedra remain, so

$$t + u - d \geq t - i, \quad d \leq i + u.$$

Only the i initially eligible tetrahedra and all newly born tetrahedra need freely chosen commitments. Their number is

$$i + 3u + 2d \leq 3i + 5u \leq 3i + 5U. \quad (4)$$

The proof of Theorem 3.3 preserves the identities of consumed original tetrahedra under every interchange, so it remains valid with the protection restriction. The sleep-set proof also preserves

this restriction. Every generated trace is supported in F , and every trace supported in F has its prescribed idle commitments on the protected originals. $\square$

The number of events in such a trace is at most $i + 2U$, even when the surrounding triangulation is much larger. The code checks the two inequalities in this proof at every search frame. A bad bookkeeping update therefore cannot silently invalidate the claimed measure.

4.2 Counting and enumerating regions

The maximum number of distinct neighbours of a vertex of G_T is four. Loops and repeated face adjacencies do not increase this bound.

Lemma 4.3 (Counting connected regions). *In a graph with $t \geq 1$ vertices and maximum degree at most four, the number of connected i -vertex subsets is at most $t16^{i-1}$ for $i \geq 1$. The number of subsets of size at most r with at most c connected components, including the empty set, is at most*

$$(r + 1)(c + 1)t^c 32^r. \quad (5)$$

Proof. Root a connected set at its least-labelled vertex. Choose a deterministic spanning tree of the induced graph and traverse its edges depth first, each twice. Record the root and the chosen local neighbour port at each of the $2(i - 1)$ steps. There are at most four choices at each step. The walk reconstructs the visited vertex set, so the encoding is injective on the connected sets selected by the deterministic rule. This proves $t4^{2(i-1)}$.

For a set of size i with $j \geq 1$ components, order the components by their least vertex. Encode their j roots, their positive sizes summing to i , and their walks. This gives the upper bound

$$t^j \binom{i-1}{j-1} 16^{i-j} \leq t^j 32^i.$$

Summing over $1 \leq i \leq r$ and $1 \leq j \leq \min(c, i)$, then adding the empty set, proves the intentionally coarse uniform bound (5). It also holds for $r = 0$ or $c = 0$. $\square$

Counting alone would not give a useful algorithm if finding those regions required enumerating all t^r subsets. The artifact uses the following explicit reverse search.

Lemma 4.4 (Canonical parent and complete reverse search). *For a connected set A with root $\rho = \min A$ and $|A| > 1$, delete the largest $x \in A \setminus \{\rho\}$ for which $G_T[A \setminus \{x\}]$ stays connected. This defines a unique parent. Starting from every singleton, and accepting an added frontier vertex precisely when this rule returns the current set as parent, enumerates each connected region once, with polynomial overhead in its size.*

Proof. A spanning tree of $G_T[A]$ has a leaf other than ρ . Deleting that leaf leaves a spanning tree on the other vertices, hence preserves connectivity of the induced graph. At least one eligible deletion therefore exists. The largest-label rule makes it unique. Iterating the parent decreases size and preserves the root, eventually reaching $\{\rho\}$.

Any child of A adds a vertex adjacent to A : otherwise it is not connected. The algorithm therefore considers it among the frontier vertices and accepts it at its unique parent. It cannot be accepted at another parent. There are at most $4|A|$ distinct frontier candidates; each parent computation needs at most $|A|$ connectivity tests on at most $|A| + 1$ vertices. This is polynomial overhead, with no enumeration of vertex-addition permutations. $\square$

For several components, the code selects connected pieces in increasing order of their least vertex. After choosing a component, it excludes its closed neighbourhood from later components. The pieces are consequently the actual components of the permitted region. Every allowed region has exactly one such ordered decomposition. The same reverse-search argument works in the remaining induced graph. An explicit stack avoids a recursion limit tied to the number of vertices.

4.3 The constructive locality theorem

Theorem 4.5 (Complete local reachability). *Let (T, h) be a valid source with $t \geq 1$ tetrahedra and initial cocycle bit bound B . Suppose a chosen endpoint predicate is invariant under the tracked marked isomorphisms and has polynomial bit complexity. Then $\mathcal{R}_{r,c,U}(T, h)$ can be searched exhaustively in*

$$t^c 2^{O(r+U)} \text{poly}(t + U + B) \quad (6)$$

bit operations. A precise sufficient counting factor is

$$(r + 1)(c + 1)t^c 32^r 11^{3r+5U}.$$

Every successful endpoint is geometrically replayable from the supplied source. If the endpoint predicate has a different worst-case cost, that cost must be included as an additional factor.

Proof. Enumerate the regions by Lemma 4.4 and its component extension. For each region use Theorem 4.2. The union is exactly the defined family: membership requires containment in at least one such region. Repeated endpoints across different regions do not affect correctness or the upper bound. Lemma 4.3 bounds the number of regions. The polynomial marking and geometric costs are established in Section 5. Multiplying these factors proves the claim. $\square$

Corollary 4.6 (Polynomial and quasi-polynomial classes). *For one connected allowed region, $r + U = O(\log t)$ gives a polynomial search bound, and $r + U = O(\log^2 t)$ gives $t^{O(\log t)} \text{poly}(B)$. More generally, $c = O(\log t)$ together with $r + U = O(\log^2 t)$ gives the latter quasi-polynomial bound.*

Proof. Take logarithms of the non-polynomial factor in (6): it is $O(c \log t + r + U)$. Substitute the stated conditions. The fixed polynomial factors do not change these classifications. $\square$

This improves what a direct move-string analysis provides for the same locality conditions. A trace has at most $r + 2U$ events but can choose among $O(t + U)$ locations at each event. That direct bound pays $O((r + U) \log(t + U))$ in the exponent. The region theorem pays $O(c \log t + r + U)$, isolating the cost of choosing the spatial components. The improvement concerns the explicit search model, not a lower bound on every alternative way to recognize knots.

4.4 A precise remaining global obligation

Suppose an outer, source-certified algorithm makes $Q(n)$ calls of this kind on sources with parameters t_i, B_i, r_i, c_i, U_i . Its accumulated cost is bounded by the sum of the individual costs, plus its other work:

$$\sum_{i=1}^{Q(n)} t_i^{c_i} 2^{O(r_i+U_i)} \text{poly}(t_i + U_i + B_i). \quad (7)$$

Thus polynomially bounded source sizes and markings, a quasi-polynomial number of outer calls, and the bounds in Corollary 4.6 would yield a quasi-polynomial total. This is a sufficient accounting

statement. It does not prove completeness of such an outer algorithm or bound its restarts. In particular, the existence of a short successful trace for an unknot alone says nothing about how a negative instance is decided.

The outstanding geometric target is now concrete: establish a useful small-region coverage theorem for a specified source construction and surface family, including its behavior after certified reductions. The existing coherent-height obstruction shows that coverage cannot be asserted without allowing changes of triangulation or a richer surface representation.

5 Bit complexity and the verification boundary

5.1 Binary cocycle values stay controlled

Lemma 5.1 (Height growth under a bounded number of expansions). *Let D_0 be the maximum absolute source edge value. After any trace with u upward events, every signed edge value has absolute value at most $2^u D_0$. A normalized tetrahedral height row and every local coherent normal coordinate have bit length at most $B + u + O(1)$ when $D_0 < 2^B$.*

Proof. A downward replacement removes its central edge and creates no new global edge. Surviving edge values are unchanged. For an upward replacement, the new diagonal joins the formal apices a, b . Choose any belt vertex v . Its value is

$$h_b - h_a = (h_b - h_v) + (h_v - h_a),$$

the sum of two old signed edge values. Its magnitude is at most twice the previous maximum. All other values persist. Induction proves the bound. Each normalized local height is an edge difference from the first corner. The coherent normal coordinates are gaps between ordered local heights, and each gap is bounded by the local span, itself an edge difference. $\square$

This bound charges arithmetic to the number of expansions, not to the potentially enormous number of normal discs. Multiplying the initial heights by a large integer changes the arithmetic bit size without requiring expanded normal sheets. The tests include a 2,048-bit marking control.

5.2 Move and identity costs

There are at most $6(t + U)$ local edge occurrences and $2(t + U)$ paired faces. Grouping edge occurrences and checking the fixed formal patterns enumerates the possible events using polynomial work. Each accepted transition uses the existing replacement producer and its independent relative-boundary checker, followed by independent cochain transport replay. The implementations may rebuild global incidence data; they are not claimed to perform every move in constant time.

An output incarnation has a structural name consisting of its event's consumed identities and local ports, its event kind, and its output slot. Repeatedly printing a fully expanded ancestry expression could be very large. The code instead interns these expressions as a directed acyclic graph with exact tuple equality. A path creates only $O(t + r + U)$ name nodes. Integer IDs are handles into the shared table; a cryptographic digest is not used as a proof that two names are equal. For the mathematical encoding argument one may fix any total order on the exact names. Within an execution, existing interned IDs never change.

If every intermediate triangulation and height list is stored in a positive certificate, there are at most $r + 2U$ moves and each list has at most $t + U$ rows. A conservative bit-size bound is therefore

$$O((r + 2U + 1)(t + U)(B + U + \log(t + U + 1))), \quad (8)$$

plus the terminal component certificate. The geometric-trace portion is polynomial in the parameters of one trace; the terminal component certificate is an additional term. The artifact uses explicit snapshots for review; it makes no unsupported claim that all certificates of an exhaustive search fit in one small proof. Storing every endpoint is optional.

5.3 Independent endpoint replay

The new endpoint certificate records the total upward allowance, the allowed initial region, the ordered geometric transition certificates, and the final normal coordinates. The verifier starts from the independently supplied source triangulation and source heights. It replays each formal replacement and cocycle transport using the existing checkers, tracks which original tetrahedra survive, counts upward events, and reconstructs the final local coordinates. It does not invoke commitment search, sleep sets, event generation, the cocycle seed solver, or a move producer.

The initial-region claim is checked by tracking original incarnations through the canonical survivor order in every recorded transition. A trace that consumes a protected original tetrahedron is rejected even if its final triangulation and surface would otherwise be valid. A false upward budget is likewise rejected. The search’s optimization bookkeeping is not trusted as a substitute for these checks.

5.4 Disc components and diagram provenance

For a supplied endpoint surface, the existing compressed normal-component kernel determines whether it contains an essential compressing disc, retaining a replayable certificate. Binary-coordinate component computations build on the interval-orbit approach of Agol–Hass–Thurston [1] and the project’s later checked implementation. The local search theorem can use any polynomial endpoint predicate; if a selected implementation has a different proved complexity, that cost must be charged separately.

No inference is made from total Euler characteristic alone. A disconnected surface can contain spheres that raise its total Euler characteristic while leaving its other components unsuitable. Likewise, preserving a cohomology class does not mean preserving the topology of its coherent fibre. The component predicate is the relevant final test.

The new `pachner_seed_decide` adapter authenticates an original knot diagram using the maintained diagram-to-exterior certificate. It optionally replays boundary shellings, constructs the starting integral cocycle, runs the bounded search, and converts a successful endpoint into the unchanged `diagram-transport-disc-v1` format. The existing independent `verify_transport_disk_certificate` checks the entire chain. A verified compressing disc in the authenticated knot exterior establishes the positive unknot conclusion through the established geometric criterion.

The statuses have the following meanings:

Status	Exact meaning
DISC_FOUND	A source-bound endpoint contains a verified compressing-disc component
ENDPOINT_SELECTED	The caller selected an endpoint; no extra topological claim is implied
COMPLETE_BOUNDED_FAMILY	The declared single-region or unrestricted finite family was exhausted
COMPLETE_BOUNDED_REGIONS	Every permitted initial region and its finite family was exhausted
INCONCLUSIVE	A required search or component query was capped, or the diagram adapter found no verified disc
UNKNOT	The diagram adapter's complete source-to-disc certificate independently replayed

The complete-family statuses are search results, not independently supplied negative unknot certificates. In particular, the diagram adapter maps bounded family exhaustion to **INCONCLUSIVE**. A caller can then invoke the maintained complete recognizer under its own remaining budget.

5.5 Resource limits and observational guarantees

The node allowance counts popped search frames. Commitment assignment frames and geometric trace frames are different quantities, so their node caps are not equal units of geometric coverage. The work allowance is a cooperative guard counter shared by preparation, moves, queries and positive replay; it is not a bit-operation counter. Disc-query caps apply at individual endpoints, and any incomplete required disc query prevents a final claim of exhausted disc search.

Callbacks receive private certificate copies. Mutation cannot alter the search state. Caller exceptions are shielded through internal cap and malformed-certificate handlers and are re-raised with their original identity. These details matter for faithful integration into an existing recognition budget.

6 Exact witness screening for normal-surface propagation

The second implementation addresses a repeated algebraic cost already identified in the incoming dual-certificate report. It is independent of Pachner search and reuses exactly the baseline's positive and negative certificate formats.

6.1 The query and its exact evidence

Fix an integer matrix $A \in \mathbb{Z}^{m \times q}$ and an integer Euler covector $c \in \mathbb{Z}^q$. For allowed columns $J \subseteq \{1, \dots, q\}$ write

$$K(J) = \{x \in \mathbb{R}^q : x \geq 0, Ax = 0, \text{supp}(x) \subseteq J, c^T x > 0\}.$$

The actual model uses standard normal coordinates, including fixed zero triangle anchors and forbidden quadrilateral types. A coordinate j is mandatory at J when every member of $K(J)$ has $x_j > 0$. The baseline tests this by solving the positive query with j deleted. A negative answer is justified by its exact Farkas multiplier and forces incompatible quadrilateral peers to zero.

Lemma 6.1 (A feasible witness excludes mandatory zeroes). *If $w \in K(J)$ and $w_j = 0$, then $K(J \setminus \{j\})$ is nonempty. More generally, a previously verified positive kernel vector w proves nonemptiness of any $K(J')$ containing its full support.*

Proof. The same vector w satisfies every defining equation and inequality of the more restricted query. Its positive objective is unchanged. $\square$

The apparent simplicity is operationally useful: a current LP point often has many zero quadrilateral coordinates, and the baseline was testing their deletion again from a cold simplex basis. A known positive point already answers all of those feasibility questions.

Proposition 6.2 (Support-antichain dominance). *If two verified positive kernel vectors have supports $S_1 \subseteq S_2$, every zero-coordinate query answered by the second is answered by the first. It is therefore sound to retain only inclusion-minimal supports. Arbitrary eviction from this collection can lose acceleration but cannot invalidate an answer.*

Proof. Any allowed set containing S_2 contains S_1 . Apply Lemma 6.1. Eviction removes evidence and never creates it. $\square$

The cache checks every candidate integer vector against the original full matrix and covector, not merely against a saved support mask. It then uses exact integer bit masks for support inclusion. Reuse across anchors and sibling branches checks the entire new column mask. A zero at the tested coordinate alone would not establish that the witness satisfies the new triangle anchors or previous branch restrictions.

Proposition 6.3 (Sparse cones can have exponentially many minimal supports). *For every $k \geq 1$, there is a positive-kernel query in $2k+1$ variables, with k equations and only coefficients in $\{-1, 0, 1\}$, having exactly 2^k inclusion-minimal positive supports.*

Proof. Use nonnegative variables $z, a_1, b_1, \dots, a_k, b_k$, equations $a_i + b_i = z$, and positive objective z . Every positive vector has $z > 0$ and at least one positive member of each pair $\{a_i, b_i\}$. Conversely, choosing exactly one member of every pair and assigning it value one, with $z = 1$ and all unchosen coordinates zero, gives a feasible positive vector. Its support is minimal. A support containing both members of any pair contains one of these smaller feasible supports and is not minimal. The 2^k choices give exactly the stated antichain. $\square$

This is an algebraic example, not a normal-triangulation realization. It shows that sparse unit coefficients alone cannot justify a polynomial unlimited-cache claim.

6.2 Preserving the exact baseline policy

The new producer always solves the current-node LP afresh using the unchanged ordered columns and unchanged exact engine. It keeps that witness separately for the current propagation sweep. It screens only coordinate deletion queries. If no known witness answers one of them, it calls the original solver. Tetrahedra and types are scanned in the same order, and the first proved mandatory coordinate is propagated as before.

Theorem 6.4 (Certificate and budget preservation). *With the deterministic LP engine and fixed baseline branch policy, every completed baseline run is reproduced by the screened producer with exactly the same positive or negative certificate. This remains true with identical finite node, branch-depth and pivot allowances, provided there is no external wall-clock or callback cutoff. The screened run spends no more simplex pivots before any corresponding unskipped query. A capped baseline may become complete after screening.*

Proof. Induct over the common logical execution. The current-node LP has the same matrix, objective and column order, so whenever it completes its deterministic answer agrees. Every omitted deletion query is positive by Lemma 6.1; it could not have supplied a negative implication or changed the first mandatory coordinate. All remaining deletion queries are solved on the same data. Hence the mandatory step, next node or branch, selected conflict, and child order agree. The positive certificates contain the same final coordinates. The negative certificates contain the same mandatory and terminal multipliers and the same exhaustive branch structure. Positive answers to the omitted deletion queries are not included in either final proof, so their omission does not change the serialized proof data.

Before any corresponding unskipped query, only some nonnegative pivot costs have been removed. The remaining pivot allowance in the screened run is therefore at least the baseline's. A query completed by the baseline cannot be cut off earlier by the screened budget. The logical node and branch-depth counts agree along this common execution. Thus any baseline completion is preserved even under those finite allowances. An external time cutoff is excluded because exact cache validation and support comparisons add work outside the simplex pivot counter. $\square$

Corollary 6.5 (Support-sensitive sweep cost). *If the current witness has s positive unresolved quadrilateral coordinates, one propagation sweep uses at most s deletion LPs. This holds even when global cache capacity is zero.*

Proof. Every other deletion is answered by the pinned current witness. The scan visits each unresolved coordinate at most once before it stops or restarts. Additional valid cached witnesses can only omit more queries. $\square$

There is no claimed polynomial bound on an unlimited antichain. The default capacity is 256 vectors, and a smaller finite capacity is supported. With capacity H , cache validation and lookup require polynomial overhead in H , the source matrix dimensions and witness bit lengths. Setting $H = 0$ retains the local guarantee while eliminating persistent antichain storage. The unchanged Bland simplex remains an exact finite algorithm without a proved polynomial worst-case pivot bound.

6.3 A rigorous quadratic-to-linear LP-call separation

The next family isolates the cause of redundant calls. It is an algebraic control with seven-coordinate blocks, not a claim that the matrices are realized by triangulations.

For $s \geq 1$, use s irrelevant prefix blocks, s mandatory blocks, and $\lceil s/4 \rceil$ auxiliary blocks. Let the first two quadrilateral coordinates of mandatory block j be $q_{j,0}, q_{j,1}$, and put $z = q_{0,0}$. Distinct triangle coordinates of the later auxiliary blocks are p_j . Impose

$$q_{j,0} = z \quad (1 \leq j < s), \quad q_{j,1} + p_j = z \quad (0 \leq j < s), \quad (9)$$

and maximize the positive query with objective z . All remaining columns and objective coefficients are zero. Every p_j has a larger original column index than $q_{j,1}$. The test harness explicitly substitutes this model and uses a matching algebraic final analysis; its proof is not submitted to the geometric normal-surface verifier.

Lemma 6.6 (Earlier duplicate columns under Bland's rule). *In the specified all-slack Bland implementation, if two original nonnegative variables have identical constraint and objective columns,*

the later variable does not enter while the earlier variable is still available and started nonbasic with it.

Proof. Every pivot applies the same linear row operations to both columns. Their tableau columns and reduced costs therefore agree. While both are nonbasic, Bland’s entering rule selects the earlier one whenever either could enter. If the earlier variable is basic, its reduced cost is zero, as is that of its twin. If it later leaves, the equality remains valid. Induction over pivots proves that the later twin never enters in the stated period. $\square$

Theorem 6.7 (Exact oracle-call counts). *For the family (9), with the specified block ordering and uncapped baseline solver, the baseline and screened propagation producers make respectively*

$$L_{\text{base}}(s) = 3s^2 + 2s + 1, \quad L_{\text{screened}}(s) = 2s + 1 \quad (10)$$

LP calls and return the same final algebraic certificate. Global cache capacity zero suffices for the screened count.

Proof. The columns $q_{j,1}$ and p_j agree, including the normalization row used by the exact solver. By Lemma 6.6, an available $q_{j,1}$ keeps p_j nonbasic and zero. Every positive current witness has $z > 0$, so each unresolved mandatory block has $q_{j,0} = q_{j,1} = z > 0$ and contains a quadrilateral conflict.

At each of the s propagation stages, the baseline tests the $3s$ irrelevant prefix quadrilateral coordinates. Their deletion leaves the positive query feasible. It then reaches type zero in the first unresolved mandatory block. Deleting this coordinate forces $z = 0$, so the test is negative and that type is propagated. Its incompatible type-one coordinate is then removed, allowing p_j to replace it in later positive points. After all s stages, the current witness is admissible. There are $s + 1$ current-node LPs and $s(3s + 1)$ deletion LPs, proving the first formula.

The irrelevant coordinates are zero in each current basic witness: they have zero objective and zero matching columns and never acquire a positive reduced cost. Screening omits their $3s$ tests at every stage. It retains the s genuinely negative mandatory tests and the $s + 1$ current-node queries. This proves the second formula using only the pinned current witness. Theorem 6.4 proves certificate equality. $\square$

Appendix A strengthens this result for the specified engine: Theorem A.1 proves exact total pivot counts $6s^3 + 3s^2 + s$ and $3s^2 + s$. These oracle and pivot separations do not by themselves establish the bit cost of every exact tableau operation or a normal-surface realizability theorem for this family.

7 Implementation and integration

7.1 An additive research path

Five new runtime modules implement the results. The existing recognizer, the exact simplex engine, the earlier propagation producer, and the geometric certificate checkers retain their pinned bytes. This makes the comparison auditable and permits callers to adopt either improvement separately.

Module in <code>fastunknot/</code>	Responsibility
<code>pachner_commitments.py</code>	Shared legal-event engine; naive, commitment and sleep-set searches; exact identity tracking
<code>pachner_commitments_verify.py</code>	Independent endpoint replay, upward count and protected-original checks
<code>pachner_regions.py</code>	Canonical-parent region enumeration and aggregate bounded search
<code>normal_pachner_search.py</code>	Diagram-to-exterior preparation and unchanged source-bound disc certificate
<code>normal_propagation_cached.py</code>	Exact local-witness and bounded antichain screening of deletion LPs

The maintained `fast/README.md` is preserved. Each new research driver has its own documentation. The dependency layer contains only additions from the incoming archives that are absent from the pinned fast tree. The new overlay is separate from that layer; every added file has a digest and a declared origin. The full reproduction snapshot also includes historical reference modules required by the inherited regression suite.

7.2 Exact geometric event representation

An event key is a tuple containing its direction and the sorted pairs (`incarnation_id`, `local_port`) for all its consumed cells. The local port is an edge number for a downward event and a face number for an upward event. Multiple local occurrences of the same global edge are reduced to one candidate. A transition is executed only after the existing formal move producer and the independent move and marking checkers accept its candidate.

The canonical local frame of the producer is anchored at the least persistent identity in its support, rather than at the least current row number. This is necessary: executing a disjoint move can change row numbers and otherwise change the new cells' corner labels. A birth key records the exact parent event and an output slot. The shared interning table gives equal birth keys equal handles across different commuting search histories. Tuple equality, not a bare hash value, resolves equality.

For endpoint-query deduplication, a state is serialized under persistent incarnation order with the corresponding normalized heights and face maps. This removes the representation changes that the proved commutations cause. It is not a general graph-isomorphism canonicalizer. Distinct, noncommuting histories can remain distinct even when their underlying unmarked triangulations are isomorphic. The current theorem does not require finding or merging all of those coincidences.

7.3 The three search methods

The naive method supplies a finite completeness oracle on tiny examples. It explores every legal move continuation, including different orders of independent moves. It uses the same geometric event engine and the same endpoint checker as the optimized methods, so measured differences isolate enumeration policy rather than a replacement of verification by trust.

The commitment method follows the proof directly. In particular, it branches over local symbols even when many assignments will never enable a move. It is useful for checking the encoding construction and for detecting accidental dependence on downward confluence. Its large constant makes it a poor default on the tested examples.

The sleep method performs geometric transitions only. Its sleep updates use exact consumed identities, and its endpoint table suppresses repeated predicate calls without suppressing continuation

expansion. It is the default of the region and diagram wrappers. All three methods use the same total-upward measure, source protection and callback boundary.

The region wrapper applies one aggregate node and cooperative-work allowance across all selected initial regions. The empty region is included and checks the source endpoint. In the current implementation, the wrapper may repeat that endpoint check in several region searches; the theorem already pays for each region. Sharing a source-independent endpoint cache across region searches is a possible further optimization, provided it does not merge continuations with different permissions.

7.4 Representative usage

All examples below run from the packaged `repro/fast/` directory or from an integrated `Topology/UnknotRecognition/fast/` directory. An unrestricted bounded search on a supplied, validated source is:

```
from fastunknot.pachner_commitments import search_pachner_endpoints
from fastunknot.pachner_commitments_verify import inspect_pachner_endpoint

answer = search_pachner_endpoints(
    triangulation, heights, method="sleep", max_upward=2,
    max_nodes=100000, seek_disc=True)
if answer["status"] == "DISC_FOUND":
    assert inspect_pachner_endpoint(
        triangulation, heights, answer["certificate"]) is not None
```

A connected-region search substitutes:

```
from fastunknot.pachner_regions import search_pachner_regions

answer = search_pachner_regions(
    triangulation, heights, max_region_size=8, max_components=1,
    max_upward=2, method="sleep", max_nodes=100000,
    max_work=5000000, seek_disc=True)
```

For diagram input, use the provenance-preserving adapter:

```
from fastunknot.diagram import Diagram
from fastunknot.normal_pachner_search import pachner_seed_decide
from fastunknot.normal_transport_verify import (
    verify_transport_disk_certificate,
)

diagram = Diagram.from_json({"braid": {"strands": 2, "word": [1]}})
answer = pachner_seed_decide(
    diagram, shellings=True, max_upward=0, max_nodes=10000)
if answer["status"] == "UNKNOT":
    assert verify_transport_disk_certificate(
        diagram, answer["certificate"])
```

The algebraic improvement has the same producer signature as its baseline, with a cache-capacity option:

```

from fastunknot.normal_propagation_cached import search_positive_euler
from fastunknot.normal_propagation_verify import (
    verify_normal_propagation_certificate,
)

answer = search_positive_euler(
    triangulation, max_pivots=100000, max_cached_witnesses=256)
if "certificate" in answer:
    assert verify_normal_propagation_certificate(
        triangulation, answer["certificate"])

```

The positive-Euler propagation statuses retain their algebraic-normal-surface scope. They do not establish that the supplied triangulation is the exterior of a particular knot diagram. This distinction remains visible in the APIs.

7.5 Policy compatibility and maintained costs

The cached propagation implementation is an additive fork because changing the baseline would invalidate the paired comparison. Its construction script requires the exact SHA-256 of the original producer before applying its specific transformations. Runtime use does not execute that script. A future maintained integration can factor the common policy into a shared internal engine after separately verifying the resulting behavior.

Witness reuse preserves the existing completed proof bytes; geometric search instead expands the set of positive proofs the caller may discover. Neither changes the default recognition pipeline in this artifact. The package therefore does not report a speedup for the entire pre-existing recognizer. It reports the measured cost of the newly exposed search and propagation operations, together with a tested optional source-bound entry point.

The main implementation costs still include rebuilding incidence structures, exact rational node LPs, terminal component analysis, and certificate copying. The work and pivot counters deliberately remain separate. A smaller pivot count need not reduce Python overhead on a tiny instance, and a smaller number of geometric traces does not make each transition constant-time.

8 Reproducible experiments and negative controls

8.1 Protocol and interpretation

The raw evidence contains the complete tested face pairings, cocycles, caps, operation counters, completed certificates and independent replay results. Experiments used Python 3.12.14; the optional external topology audit used Regina 7.4. Timings were collected in a shared execution environment without CPU isolation. They are descriptive measurements on the declared examples, not estimates of an asymptotic exponent or population-wide performance.

The geometric benchmark has one warm-up pair and three recorded pairs per size, with seeded randomized arm order. It reports the median of the three within-pair naive/sleep ratios. The LP audit has three alternating-order pairs per case and reports the ratio of the two arm medians. These statistics are intentionally named separately: they need not give the same number. Neither protocol warrants narrow confidence intervals from three repetitions. Raw observations allow a later isolated rerun and a different statistical analysis without reconstructing the sources.

8.2 Finite completeness and independently checked geometry

The geometric audit compares naive, commitment and sleep exploration on eleven small marked instances, including bounded upward search, relabelling, protected initial cells and a 2,048-bit marking. Together with the maintained coherent obstruction, it records 34 search calls. There are no endpoint-set mismatches among the three methods on the finite comparisons.

An independent endpoint checker accepts 91 audit certificates; 90 targeted mutations are rejected. These include changes that would falsify source protection or the upward allowance, in addition to geometric and marking corruptions. A separate replay of the geometric timing file accepts all 126 saved endpoint certificates. The benchmark timings exclude this later external replay, but include transition validation and certificate copying performed by the search itself.

The optional Regina audit examines 23 distinct triangulation states and 176 replacements. It checks the eligible move inventory and the isomorphism signature of each resulting triangulation. No inventory or replacement mismatch is observed. This is an independent finite check of the repository’s formal move interpretation; it does not replace the general relative-boundary proof or certify an untested parameter range.

The new focused tests also compare the canonical-parent region enumerator with powerset enumeration on every labelled simple graph with at most four vertices and on three structured six-vertex graphs, across all relevant size and component caps. They test loops, duplicate adjacencies, validation, cancellation and callback exception identity. The diagram adapter tests source mismatch and corrupted proof rejection with the existing checker. The additional region-and-diagram audit supplies executable small examples of the union-of-regions semantics: five cases are compared with 120 literal allowed-subset searches, with 302 successful endpoint replays and no endpoint or aggregate-count mismatch. Nine diagram calls yield four independently replayed positive proofs, and three wrong-source substitutions are rejected. The four positives use two diagram sources under different options and have zero Pachner moves. They validate the source adapter, not additional discovery through moves.

There is also an explicit limited-family miss. The two-strand closure of the word $[1, -1, 1]$ is an unknot, since the first two generators cancel. With boundary shelling enabled and the allowed initial region empty, the adapter exhausts its declared region family and returns **INCONCLUSIVE**. This small control makes the coverage restriction observable in executable data. It is not a lower bound for nonempty regions.

8.3 Independent collapse regions: a geometric control

For each tested $p = 1, \dots, 6$, the driver starts with a layered solid torus on $2p$ tetrahedra and applies 2–3 moves in p disjoint pairs. The result has $3p$ tetrahedra and carries the transported cocycle. Its tested downward event inventory consists of the p independent inverse moves; the audit also checks that no additional downward event appears at later endpoints. This realizes the hypotheses of the operation-count calculation in Section 3 at each reported size.

At $p = 6$, the unreduced search visits 1,957 trace nodes and sleep search visits 64. The median recorded times are 9.225 and 0.416 seconds, and the median paired speedup is 22.18. The gain grows with the repeated-order work removed in this control. At $p = 1$, both methods visit two nodes and sleep bookkeeping makes the measured time slightly worse. This regression is retained rather than removed from the comparison.

The distinction between the proof algorithm and the practical algorithm is large. On the $p = 2$

Table 1: Genuine independent-collapse controls. Times are medians in milliseconds; the final column is the median within-pair ratio.

p	t	N_{naive}	N_{sleep}	Naive (ms)	Sleep (ms)	Paired ratio
1	3	2	2	1.273	1.453	0.876
2	6	5	4	13.195	10.386	1.270
3	9	16	8	40.916	22.266	1.838
4	12	65	16	212.168	59.087	3.593
5	15	326	32	1,260.972	158.664	8.346
6	18	1,957	64	9,224.509	415.842	22.183

One warm-up pair is excluded. Three measured pairs per size; seeded randomized arm order. Full sources and certificates are saved.

Table 2: Exact LP screening on genuine triangulations. Each paired entry is baseline/screened; times are seconds.

Source	LP calls	Pivots	Median seconds	Ratio	Result
Layered, 1 tet.	3/3	7/7	0.001/0.001	0.927	+
Layered, 2 tet.	7/6	59/50	0.018/0.016	1.171	+
Layered, 4 tet.	15/10	354/264	0.356/0.267	1.332	+
Layered, 6 tet.	23/14	1,073/784	2.105/1.559	1.350	+
Layered, 8 tet.	31/18	2,620/1,948	9.682/8.151	1.188	+
Branching torus	41/22	1,134/648	1.684/0.908	1.855	+
Finite trefoil	20/20	1,554/1,554	2.397/2.445	0.981	0
Figure-eight, frozen	2/2	600/600	3.872/4.054	0.955	cap
Figure-eight, relabelled	2/2	600/600	3.976/3.995	0.995	cap
Figure-eight, native	1/1	600/600	4.358/4.121	1.057	cap

+: completed positive-Euler query; 0: completed negative positive-Euler query; cap: inconclusive at 600 pivots. Three alternating-order pairs per case; ratio of arm medians. The result concerns the stated surface query, not an unauthenticated knot-diagram verdict.

control, full commitment enumeration visits 175,905 assignment frames, whereas the sleep search has four trace nodes. Restricting the permitted originals to one three-tetrahedron inverse-move region reduces the commitment search to 401 frames. These are different units of work, but they show why the constant-alphabet encoding should not itself be sold as a fast practical traversal. The sleep method inherits its bound without paying for most inert assignments.

The maintained eight-tetrahedron coherent obstruction supplies a different control. Sleep exploration finds an independently verified disc after one downward move and two endpoint queries. This confirms that the implemented search can escape that particular fixed-triangulation coherent obstruction. It establishes neither a worst-case locality bound nor a robust timing gain over the incoming greedy transport code.

8.4 Exact LP reuse on genuine triangulations

The LP audit contains fifteen cases and 45 paired samples, hence 90 arm runs. Each arm has a 35-second wall allowance. The figure-eight cases also have a 600-pivot allowance. No arm hits the wall allowance. Of the 45 pairs, 36 complete on both sides and have byte-equal certificates. All 21 completed genuine-triangulation pairs independently verify in both arms. The other 15 completed pairs are the explicitly substituted algebraic controls.

Table 3: Algebraic controls, with no asserted triangulation realization. Every call and pivot total agrees with the proved formula.

s	LP calls (base/screen)	Pivots (base/screen)	Median seconds	Ratio
1	6/3	10/4	0.002/0.001	1.756
2	17/5	62/14	0.018/0.005	3.732
4	57/9	436/52	0.411/0.053	7.781
6	121/13	1,410/114	2.677/0.249	10.771
8	209/17	3,272/200	10.793/0.706	15.289

Both producers complete with the same substituted algebraic certificate. Ratios are baseline median divided by screened median.

The branching solid torus is the strongest genuine example in this small cohort. Calls fall from 41 to 22, pivots from 1,134 to 648, and median elapsed time from 1.684 to 0.908 seconds. Fifteen omissions come from the pinned current witness and four from other verified antichain witnesses. This shows that cross-query reuse can help beyond local zero support.

The layered family gives smaller improvements. At eight tetrahedra, calls fall from 31 to 18 and pivots from 2,620 to 1,948; the median-time ratio is 1.19. The one-tetrahedron case makes the same three LP calls and seven pivots, so added validation and cache management produce a small slowdown. The finite trefoil similarly makes the same 20 calls and 1,554 pivots, with a median-time ratio of 0.98. Its completed negative certificate remains exactly equal to the baseline's. A negative result for this positive-Euler surface query should not be relabelled as a new general knot-recognition algorithm.

8.5 Figure-eight controls expose the unsolved cost

The frozen figure-eight exterior, its relabelled copy, and a regenerated native finite figure-eight exterior all remain inconclusive at 600 pivots. There are no screenable deletion queries in these runs. The frozen versions start two LPs before the allowance is consumed; the native source consumes it in its first LP. Thus the new cache does not address the current-node LP bottleneck observed here.

The native fixture is reconstructed with Regina's figure-eight example, conversion to a finite triangulation, and simplification. Its full face pairings, version and isomorphism signature are saved. Within each pair the two producers receive exactly the same reconstructed source. Fresh simplification need not produce the same combinatorial triangulation, even within the same Regina version. Reproduction therefore validates each fresh source without requiring its signature to equal the frozen source's. The saved native signature is

kLLLALkIcegfegjijjhqaxrlmlf.

Wall-time differences between two capped, inconclusive runs are not evidence of faster completed recognition. They remain in the raw data and the table to show both the resource boundary and the absence of witness hits.

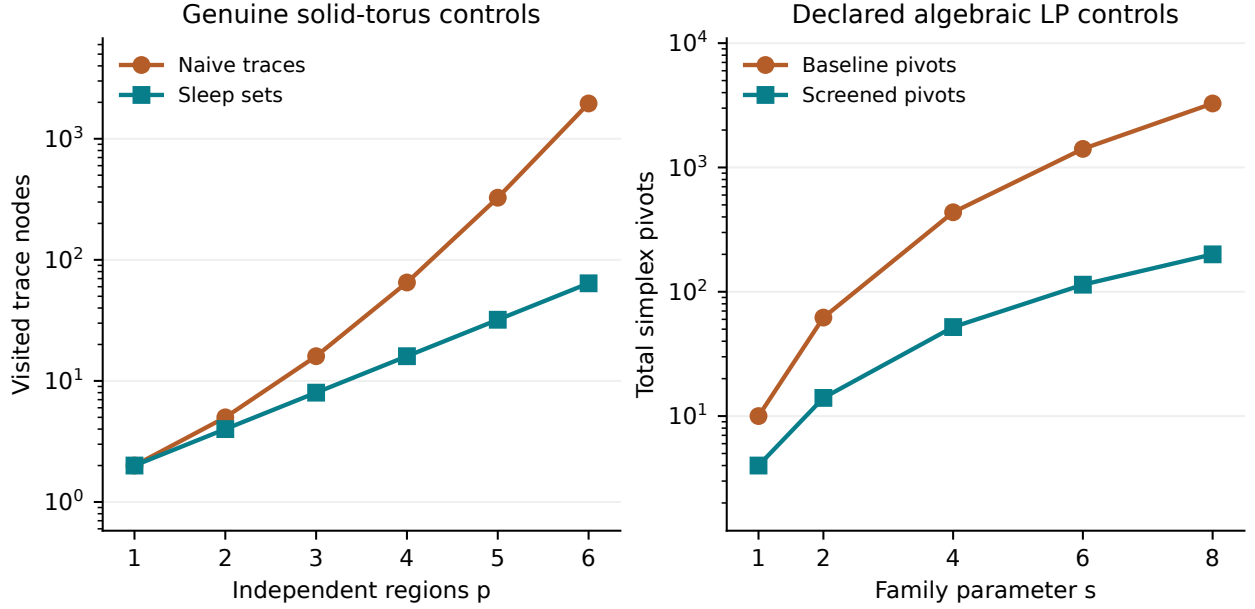


Figure 1: Measured operation counts on the two controlled families. Left: actual triangulations with independently collapsible regions, showing all prefixes before and after sleep reduction. Right: the declared algebraic LP family, showing exact total pivot counts. Logarithmic vertical axes are used only for readability. Curves connect the tested integer sizes; the formulas and their hypotheses, rather than these plots, establish the asymptotic statements.

8.6 The algebraic separation, measured and proved

Every measured call count agrees with Theorem 6.7. The exact pivot induction in Appendix A proves, for this solver and model ordering,

$$P_{\text{base}}(s) = 6s^3 + 3s^2 + s, \quad P_{\text{screened}}(s) = 3s^2 + s.$$

At $s = 8$, this is 209 versus 17 LP calls and 3,272 versus 200 pivots, with a measured median-time ratio of 15.29. This is a rigorous cubic-to-quadratic pivot separation for the algebraic family, supported by finite controls. It is not a proved cubic-to-quadratic bit-time theorem for arbitrary simplex tableaux, and it does not supply a triangulation realizing the family.

8.7 What the evidence supports

The geometric result is complete search of a declared finite move family, with independent replay and a genuine example of factorial ordering redundancy. The LP result is proof-preserving query omission, with moderate genuine-triangulation gains and a stronger isolated algebraic separation. The source adapter gives independently replayed positive diagram verdicts. None of these observations establishes that every unknot has a successful small-region trace, that every negative instance is settled by bounded coherent search, or that exact node LPs have polynomial complexity.

The inherited and added regression tests are covered by the full inventory gate described in Section 10. The archive preserves its logs, source digests and all failed or interrupted preliminary runs rather than replacing them with the final successful summary.

9 A concrete route toward quasi-polynomial recognition

The locality theorem separates a search-theoretic cost that is now controlled from geometric obligations that are not. This section states those obligations as questions with testable intermediate targets. A change in a parameter or source class should be reflected in the theorem statement, rather than hidden inside an empirical success rate.

9.1 The sufficient coverage theorem we would need

Here is one precise form of a future global result. It allows a controlled list of sources, rather than assuming that a single height seed works on every diagram.

Proposition 9.1 (Conditional recognition consequence). *Suppose there is an algorithm that, from an n -crossing knot diagram, constructs at most $n^{O(\log n)}$ authenticated exterior triangulations with specified integral cocycles and region budgets. Assume its construction cost is $n^{O(\log n)}$, all source sizes and marking bit lengths are polynomial in n , every region budget satisfies $c = O(\log n)$ and $r + U = O(\log^2 n)$, and a complete component-disc predicate has polynomial bit complexity on each resulting endpoint. Finally assume the following uniform coverage property: every unknot diagram has a compressing-disc endpoint in at least one of the resulting region families. Then exhaustive region search gives an $n^{O(\log n)}$ unknot-recognition algorithm.*

Proof. Theorem 4.5 bounds each search by $n^{O(\log n)}$; multiplying by the number of sources preserves this bound. Source authentication and component-disc replay make every positive verdict sound. The uniform coverage property guarantees a positive endpoint on every unknot. Therefore, after all the finite searches finish without one, the diagram is knotted. The stated predicate cost ensures that this exhaustion is an algorithm, not a search with unresolved endpoint queries. $\square$

The proposition identifies exactly how a positive coverage theorem would also justify negative decisions by exhaustion. No separate short negative certificate is required for that logical consequence. The current artifact does not establish uniform coverage and therefore does not take that final negative step. A few short successful traces, or a coverage theorem only for an already recognized subfamily of unknots, would be insufficient.

An alternative program can work through a certified hierarchy. In that case one must bound the number, size and marking cost of all subsequent search sources, as in (7), and prove the outer hierarchy complete. A local simplification that can restart without a global decreasing measure does not yet supply that bound.

9.2 Priority 1: geometric locality and an honest obstruction search

1. Which source families admit a logarithmic or squared-logarithmic region? Fix a concrete diagram-to-exterior construction, its shelling policy and its cocycle normalization. Ask whether every solid torus in this source family admits a disc endpoint with $r + U = O(\log^2 t)$, first with one connected allowed region and then with $O(\log t)$ components. The theorem should specify whether the cocycle is fixed, optimized by vertex potentials, or selected from a bounded list. Changing that choice changes the coverage problem.

A useful initial milestone is a proof for one explicit infinite family of unknot diagrams that survives the existing shortcut stages and creates nontrivial interior move choices. Merely testing a

family whose initial coherent seed is already a disc would establish little about geometric search. The cancelled-braid empty-region miss and the eight-tetrahedron obstruction are starting controls with different failure mechanisms.

2. Can the required region or upward budget be proved large? For increasing triangulated solid tori, compute the minimum allowed-region size and minimum total upward count separately. Certified lower bounds require complete exhaustion of the smaller family, with no capped endpoint queries. Such computations can refute an overly strong conjecture on a finite example, while a family construction could rule out a proposed uniform logarithmic theorem. Useful adversarial sources include long layered regions, separated interacting defects, and relabelled copies that expose heuristics based on arbitrary row order.

The target is not to optimize a single scalar $r + U$ before understanding both terms. A small permitted spatial region may still need many expansion events, and a trace with few upward events can consume a large initial region. The current count makes their different geometric meanings visible.

3. Can hierarchy structure produce the needed regions? Relate the initial consumption footprint to the product regions, multi-surfaces and controlled decompositions in the hierarchy program [7, 8]. A useful lemma would take one precisely specified reduction object, produce a legal marked trace, and bound its consumed original cells, number of components and total expansions. The proof must include the representation of the surface and its marking after cutting. An existence theorem for a topological simplification without this translation leaves the algorithmic parameters unbounded.

The successful reduction object may be much smaller than the normal surface it encodes. Binary multiplicity is therefore compatible with a small region theorem; enumerating all normal sheets to locate that region is not. Conversely, the height-growth lemma alone controls arithmetic and gives no spatial localization. These two estimates should be proved separately and combined only at the final cost accounting.

4. Can small excess height be converted to a small total budget? Pachner-graph experiments often constrain maximum excess tetrahedron count [2], whereas the theorem here constrains total upward events. A conversion theorem would need to remove redundant excursions or bound essential returns. Commuting disjoint moves removes order redundancy but does not eliminate a sequence of dependent expansion-collapse cycles. An exact state-based loop-removal argument may help with shortest traces; its state count and marking dependence must be included. A bound on upward bursts has the same issue and should not be substituted for U .

9.3 Priority 2: stronger complete geometric search

5. Can separators replace total active-region size? The commitment bound charges every freely assigned incarnation. In a long trace whose interactions have a small separator, a dynamic program might reuse completed subregions instead. A promising parameter is the width of the dependency graph of events, with edges whenever one event consumes a cell created or constrained by another. The first theorem needed is a bounded interface description: which boundary face maps, local cocycle values, surviving identities and surface data suffice to compose partial traces? Without that description, invoking treewidth only renames the state-explosion problem.

A concrete target is a family with $r = \Theta(t)$ but logarithmic interface width, together with a finite compatibility relation and a proved state count polynomial in the marking bit length. This would extend the useful class beyond small initial footprints.

6. Which additional local relations safely reduce trace classes? The present sleep relation uses disjoint supports. Burton’s overlapping composite relations suggest larger reductions [2]. Each candidate needs a marked relative-boundary proof and an event-identity rule that remains stable under other reductions. Test completeness against the naive oracle before evaluating speed. A raw triangulation transposition table must include remaining resources and continuation restrictions, or supply a dominance proof. Equal unmarked endpoints alone do not justify dropping one search state.

Region enumeration itself offers a modest immediate target: share exact endpoint-predicate results across allowed regions while retaining separate continuations. The proof should distinguish the terminal surface property, which can be reused, from the permissions to consume original cells, which differ. This avoids repeatedly checking the source endpoint and other common states without changing the searched union.

7. Can the framework include the other Pachner moves? For unmarked formal star replacements, a tentative alphabet would have six edge ports, four face ports, four vertex ports, one whole-cell port and idle, for sixteen symbols. If u_{23}, u_{14} count expansions and d_{32}, d_{41} count contractions, set $E = u_{23} + 3u_{14}$. The tetrahedron balance gives

$$d_{32} + 3d_{41} \leq t + E.$$

The birth count then obeys

$$t + 3u_{23} + 4u_{14} + 2d_{32} + d_{41} \leq 3t + 5E.$$

This accounting suggests an analogous constant-alphabet theorem, but the formal-star legality, commutation and marking rules must be proved before asserting it for an implementation. In particular, a 1–4 move introduces an interior vertex whose cocycle height needs a canonical choice or an explicitly bounded search. The current artifact does not implement that extension.

8. Can positive traces be made smaller without weakening replay? The explicit-snapshot certificate is polynomial for one bounded trace, but repeats unchanged surrounding tetrahedra at every move. A persistent face-map representation or a shared-prefix certificate DAG could remove that duplication. The independent verifier should accept a stream of local changes, check the same relative-boundary facts, and bound its memory. Storing all endpoints of an exponential search is a different problem: a compact proof of one success does not compress the whole search tree.

9.4 Priority 3: exact LP discovery and normal-surface structure

9. Can a genuine triangulation realize a superconstant screening gain? The algebraic separation has unit coefficients and sparse equations, but normal matching matrices also encode very specific face and disc-type incidences. Construct a family of actual compact triangulations for which the ordered baseline makes asymptotically more deletion calls than local witness screening. A weaker useful first target is a rigorous lower bound on the number of redundant baseline tests in a layered or glued family, with a matching positive point and independently checked topology. The finite solid-torus timings do not establish such an asymptotic family.

10. What geometric hypotheses bound the support antichain? Proposition 6.3 rules out a small-antichain argument from sparsity and unit coefficients alone. Possible additional hypotheses include a controlled decomposition of the triangulation, a bounded number of quad types interacting with the boundary, or a restricted rank-one cone. The desired theorem should bound the number of useful minimal supports or provide a small subcollection covering all deletion queries encountered by a specified branch policy. Bounding all supports may be stronger than the algorithm needs.

11. Can first-node LPs be accelerated with exact dual lifting? The capped figure-eight controls make this a direct implementation priority. An equality-row basis can reduce a redundant matrix before simplex. A certificate of the row transformation allows a dual of the reduced problem to be lifted to the original equations. Fraction-free elimination and sparse factorization are natural candidates, but their exact coefficient growth, per-mask rank changes and reconstruction cost need measurements and proofs.

Such an optimization can preserve mathematical soundness while changing the exact serialized multiplier and hence falls outside the byte-preserving mode proved here. It should have its own compatibility contract. More ambitiously, a polynomial-bit-complexity LP method can replace the pivot engine while retaining exact independently checked primal or dual outputs; one must count its recovery and verification work, not merely invoke a real-arithmetic LP bound. Warm starts are another practical direction, but need careful basis validation after coordinates are deleted.

12. How should positive and negative evidence interact? Positive supports safely answer any further zero restriction containing them. A negative dual has the opposite issue when columns are restored: its inequality must be checked on every newly allowed column. A two-sided cache could reuse both forms of evidence with full matrix checks. Its branch policy and final proof may change, so compare it against the present byte-preserving producer rather than silently weakening that guarantee. Useful quantities include exact validation cost, cache hit location, and the number of current-node LPs avoided.

13. Can propagation backdoors be bounded after geometric preprocessing? The incoming dual-certificate work isolates quadrilateral propagation backdoors. The locality search may expose triangulations with smaller backdoors even when it does not directly find a disc. A joint theorem would bound both the cost of reaching such a triangulation and the number of unresolved choices afterwards. The endpoint predicate would then be algebraic complexity reduction rather than immediate disc discovery. It must remain source-bound, and the outer choice of endpoints must not reintroduce an uncontrolled number of LP searches.

9.5 Priority 4: evaluation that can disprove the proposed story

14. What does a matched source-level cohort show? Create a frozen cohort stratified by original diagram crossings, exterior tetrahedron count, known shortcut success and initial coherent-disc status. Every arm should receive identical face pairings and marking choices. Compare time to a verified positive or negative result, the proportion remaining inconclusive under explicit caps, total upward moves, region size, endpoint-query time, LP pivots and certificate size. Independent relabellings and matched constructor versions are essential controls for order-sensitive heuristics. The present fifteen LP cases and six geometric control sizes are useful mechanism tests, not such a representative recognition study.

15. Which empirical claims survive fixed-resource reproduction? Repeat paired timing in an isolated environment, report all censored runs, and retain the raw per-pair observations. Separate source construction, search, terminal verification and later independent replay when their cost matters. A speedup that disappears after including required verification is not an end-to-end gain. A future benchmark should also include the unfavorable small cases and node-LP bottlenecks already observed here.

The highest-value next result is a geometric theorem bounding the parameters in Proposition 9.1, or a rigorous obstruction showing that a proposed bound fails. The next implementation result is exact current-node LP acceleration with a checked reconstruction map. Both have clear acceptance criteria and address limitations that the present proofs and experiments identify directly.

10 Reproduction, provenance and integration

10.1 Archive organization

The ZIP is designed to be both a readable research record and an additive integration proposal. Its root directory contains the report sources and compiled PDF, a runnable source snapshot, raw evidence and an integration tool. The file manifest records byte lengths and SHA-256 digests. The integration manifest additionally identifies which additions come from the earlier incoming packages and which are introduced by this continuation.

Location	Contents
<code>article/</code>	Main TeX, section sources, bibliography, generated tables, figure PDF/PNG and compiled article
<code>repro/fast/</code>	Pinned fast tree plus dependency additions and the five new runtime modules; tests and research drivers
<code>repro/reports/</code> , <code>repro/synthesis/</code>	Exact historical reference modules and fixtures needed by inherited tests
<code>integration/</code>	Dependency and new-work overlays, code-review patch, preflight/apply tool and origin manifest
<code>provenance/</code>	Pinned source digests, incoming archive digests, integration audit and complete regression logs
<code>scripts/</code> <code>README.md</code>	Evidence-table/figure regeneration and package integrity verification Entry points, dependency notes, theorem scope and reproduction commands

The source snapshot avoids a dependency on the current GitHub main branch. The external mathematical literature is cited by stable versions or precise document URLs. Full third-party papers are not copied into the archive. The three prior incoming archives remain identified by their repository paths and immutable Git objects; the executable additions needed from them are included directly.

10.2 Run the new focused tests

The new runtime code uses the Python standard library. The optional topology cross-check uses Regina. Plot regeneration additionally uses Matplotlib; building the report uses a normal L^AT_EX installation with the listed packages. Python 3.12.14 and Regina 7.4 describe the recorded execution,

not a promise that every other version gives identical timings or simplex implementation behavior after source changes.

From `repro/fast/`:

```
python -m unittest \
    tests.test_pachner_commitments \
    tests.test_pachner_regions \
    tests.test_normal_pachner_search \
    tests.test_normal_propagation_cached -v
```

These four modules contain 29 new tests. The inherited test suite is much larger and includes historical reference imports outside `fast/`; the supplied companion directories preserve those relative paths.

10.3 Replay saved evidence before rerunning timings

The following commands independently replay the saved geometric and source-bound evidence. They do not rerun the expensive benchmark search.

```
python -m commitment_research.run replay \
    --input commitment_research/results/audit.json \
    --output commitment_research/results/reproduced_audit_replay.json
python -m commitment_research.run replay \
    --input commitment_research/results/benchmark.json \
    --output commitment_research/results/reproduced_benchmark_replay.json
python -m region_research.audit \
    --replay region_research/results/audit.json \
    --output region_research/results/reproduced_replay.json
```

The LP evidence has a separate verifier-only command:

```
python -m lp_cache_research.replay \
    --output lp_cache_research/reproduced_replay.json
```

It independently checks all seven distinct completed genuine certificates, 42 paired-sample certificate-digest references and ten genuine source digests. It reconciles all fifteen summary rows, but excludes the three inconclusive genuine cases and five algebraic certificates from geometric certificate replay. No search, LP solver or native constructor is imported. The focused cache test suite repeats the positive, negative and policy controls. The algebraic harness intentionally substitutes model construction and final analysis; its certificates must not be passed off as geometric normal-surface certificates.

To reconstruct the finite audits and paired measurements:

```
python -m commitment_research.run audit \
    --output commitment_research/results/reproduced_audit.json
python -m commitment_research.run benchmark --repeats 3 \
    --output commitment_research/results/reproduced_benchmark.json
python -m region_research.audit \
    --output region_research/results/reproduced_audit.json
python -m lp_cache_research.audit --rounds 3 --timeout 35 \
    --figure-pivots 600 --output lp_cache_research/reproduced_results.json
```

The optional external move inventory check is documented in `commitment_research/README.md`. It requires Regina and compares exact source and target triangulations. Replaying a saved file

Table 4: Complete final regression gate. Each test module belongs to one selected passing batch.

Batch	Modules	Tests	Seconds	Result
1	31	227	59.898	PASS
2	30	221	38.288	PASS
3	30	280	21.664	PASS
4	30	238	35.743	PASS
5	30	256	28.164	PASS
6	30	215	17.531	PASS
Total	181	1,437	201.289	PASS

Zero failures, errors or skips in the final selected batches; zero module omissions. The preliminary interrupted run and the corrected native-fixture test failure are preserved separately.

and regenerating a constructor-dependent fixture are different tasks, and the drivers retain that distinction.

10.4 The complete regression gate

The final gate covers the complete inventory of test modules under the combined fast tree. It uses six fresh Python processes with each module assigned exactly once; the machine-readable record lists module and test identities, source digests, counts, statuses and per-batch logs. This avoids depending on a single long-lived process for a suite with many historical and optional native-library tests.

The preliminary history is retained. An initial single-process discovery run encountered omitted historical companion files and later ended without a normal unittest summary. Those files were restored byte-for-byte from the pinned revision. A focused retry of the affected modules passed. During the complete inventory run, one newly added optional test incorrectly required a freshly simplified Regina figure-eight exterior to equal a particular frozen triangulation. The test was corrected to validate each source against its own construction or saved record. The affected batch was then rerun; its initial failure and corrected source digest remain in the provenance. The producer, baseline LP engine and benchmarks were not changed to resolve that test assumption.

The passing gate is therefore a claim about a precise final source snapshot, not a claim that every preliminary run passed. Source hashes also make clear which post-measurement edits only improved documentation. The portable `scripts/rerun_regression.py` lists the exact partition by default and repeats it with `-run`, writing fresh logs without replacing the recorded evidence.

10.5 Apply the proposed additions

The integration tool preflights every target path and baseline prerequisite before creating any files. It accepts an existing addition only when its bytes are identical, refuses conflicting content and performs no replacement of a maintained file. It defaults to a review operation; the explicit apply option creates the missing additions. From the archive root:

```
python integration/apply_overlay.py --repo /path/to/ProveIt --check
python integration/apply_overlay.py --repo /path/to/ProveIt --apply
```

The pinned-revision check and the prerequisite digests prevent an apparently successful integration against incompatible source APIs. A later repository revision should be reviewed explicitly, rather

than patched by relaxing a digest mismatch. The full overlay includes reproducibility material; the code-review patch is a convenient view of Python changes, not a substitute for the dependency and evidence layers.

The packaging audit applies the additions to a clean local copy of the pinned fast source, checks all resulting file hashes, confirms that no maintained source bytes changed, and checks a second application for idempotence. It does not publish or push anything to GitHub.

10.6 Rebuild and verify the report

From the archive root, first verify the shipped bytes, then regenerate the tables and figure from the saved measurements and compile:

```
python scripts/verify_manifest.py
python scripts/build_evidence.py
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -cd article/article.tex
```

The shipped PDF was compiled with resolved citations and references, then rendered page by page for visual inspection. The generated plots use the recorded integer operation counts and clearly distinguish genuine triangulations from the algebraic control. Recompile can change PDF metadata or engine-dependent byte layout; the integrity manifest authenticates the shipped bytes, not every future rebuild of equivalent content.

A Exact pivot counts for the algebraic screening family

The oracle-call separation in Theorem 6.7 can be strengthened for the exact simplex implementation supplied with the repository. This appendix proves the pivot counts, including the entering and leaving choices; it does not infer them from the experimental table. The statement concerns the algebraic family (9), with its specified ordering and solver. No realization by three-manifold triangulations is asserted.

Theorem A.1 (Exact pivot separation). *For the s -block algebraic family of Theorem 6.7, run the specified all-slack Bland solver without a pivot cutoff, or with an allowance sufficient for both complete propagation runs. The total numbers of simplex pivots are*

$$P_{\text{base}}(s) = 6s^3 + 3s^2 + s, \quad P_{\text{screened}}(s) = 3s^2 + s. \quad (11)$$

Every positive current-node or irrelevant-coordinate deletion query uses exactly $2s$ pivots. Deleting the mandatory type-zero coordinate of block j , where $0 \leq j < s$, uses exactly $2j$ pivots. Global cache capacity zero suffices for the screened formula.

We first establish a tableau lemma that includes all the restrictions arising during these propagation runs.

A.1 Ordered chain systems and the exact initial tableau

Let $m \geq 1$. Consider disjoint groups $G_0, \dots, G_{m-1}$ of nonnegative original variables, none containing a distinguished variable z . A group may contain several variables with identical columns, or it may

be empty. Write

$$X_i = \sum_{x \in G_i} x.$$

Let $\mathcal{Y}$ contain all other original variables, whose matching columns and objective coefficients are zero, and put $Y = \sum_{y \in \mathcal{Y}} y$. The matching equations, in this exact order, and the objective are

$$X_i - z = 0 \quad (0 \leq i < m), \quad \text{objective } z. \quad (12)$$

The normalization used by the solver is

$$z + \sum_{i=0}^{m-1} X_i + Y \leq 1. \quad (13)$$

The implementation forms the inequalities $Cx \leq 0$, followed by $-Cx \leq 0$, followed by (13). It adds one nonnegative slack to each row and starts with the all-slack feasible basis. Denote the slacks of the C rows by $a_0, \dots, a_{m-1}$, those of the $-C$ rows by $b_0, \dots, b_{m-1}$, and the normalization slack by σ . Thus the initial tableau equations are

$$a_i + X_i - z = 0 \quad (0 \leq i < m), \quad (14)$$

$$b_i - X_i + z = 0 \quad (0 \leq i < m), \quad (15)$$

$$\sigma + z + \sum_{i=0}^{m-1} X_i + Y = 1. \quad (16)$$

Every original variable precedes every slack in the variable order. Slack indices then follow the displayed row order: all a_i , all b_i , then σ . Removing an original column preserves the relative order of the remaining original variables and of all the slacks.

The entering rule chooses the least-index variable with strictly positive reduced objective coefficient. The leaving rule minimizes the usual nonnegative ratio and, among equal ratios, chooses the least-index current basic variable. Zero reduced coefficients are never selected. The solver returns immediately once the objective is positive, or when no positive reduced coefficient remains.

Lemma A.2 (Exact chain pivot path). *Assume z is present. If every G_i is nonempty, the solver for (12) returns a positive point after exactly $m + 1$ pivots. If h is the first index for which G_h is empty, it returns a nonpositive certificate after exactly $h + 1$ pivots. These assertions are independent of the number of irrelevant variables and of the number of available duplicate columns in each nonempty group.*

Proof. Initially z is the only variable with positive reduced coefficient. Its coefficients are -1 in the C rows, $+1$ in the $-C$ rows, and $+1$ in the normalization row. All $-C$ rows have ratio zero, whereas the normalization row has ratio one. The leaving variable is therefore b_0 , by the specified slack ordering and Bland tie rule. The pivot coefficient is one, the objective remains zero, and the resulting dictionary expresses

$$z = X_0 - b_0.$$

Here is the full inductive invariant. After $h + 1$ pivots, suppose that $G_0, \dots, G_{h-1}$ were nonempty. Let x_i be the least-index available member of G_i that entered for $i < h$, and put $Y_i = X_i - x_i$.

The basic original variables are $z, x_0, \dots, x_{h-1}$. All a_i , all b_j with $j > h$, and σ are also basic. The nonbasic slacks are $b_0, \dots, b_h$. The tableau equations are exactly

$$z - X_h + b_h = 0, \quad (17)$$

$$x_i + Y_i - X_h + b_h - b_i = 0 \quad (i < h), \quad (18)$$

$$a_i + b_i = 0 \quad (i \leq h), \quad (19)$$

$$a_j + X_j - X_h + b_h = 0 \quad (j > h), \quad (20)$$

$$b_j - X_j + X_h - b_h = 0 \quad (j > h), \quad (21)$$

$$\sigma + (h+2)X_h - (h+1)b_h + \sum_{i < h} b_i + \sum_{j > h} X_j + Y = 1. \quad (22)$$

The current objective value is zero, and its reduced expression is

$$z = X_h - b_h. \quad (23)$$

All tableau right-hand sides except that of the normalization row are zero. Direct elimination in (14)–(16) proves the invariant for $h = 0$.

If G_h is empty, equation (23) has no positive reduced coefficient: b_h has coefficient -1 , and all other nonbasic variables have coefficient zero. The solver therefore stops after $h+1$ pivots with its nonpositive certificate.

Suppose instead that G_h is nonempty. Precisely its variables have positive reduced coefficient, each equal to one. Bland consequently selects $x_h = \min G_h$. Its coefficient is -1 in the z row and in each earlier x_i row, zero in the a_i rows with $i \leq h$, -1 in the later a_j rows, $+1$ in the later b_j rows, and $h+2$ in the normalization row. Thus the only eligible leaving rows are the later b_j rows, with ratio zero, and normalization, with ratio $1/(h+2)$.

If $h < m-1$, a zero-ratio row exists. The remaining basic variables b_j , $j > h$, retain their original indices, so Bland chooses b_{h+1} . This is a unit pivot in a row with zero right-hand side. Substitution using

$$X_h = X_{h+1} - b_{h+1} + b_h$$

gives every invariant equation with h replaced by $h+1$; the new basic variable x_h supplies the additional instance of (18). The normalization coefficients become $h+3$ and $-(h+2)$, as required. The objective is still zero. This establishes the induction, including both Bland choices.

If $h = m-1$, there is no later b_j row. The only eligible leaving row is normalization, with pivot coefficient $m+1$. This final pivot sets $x_{m-1} = 1/(m+1)$ and makes the objective $1/(m+1) > 0$. The implementation returns immediately. The initial z pivot and the m entering-group pivots give exactly $m+1$ pivots.

Throughout the zero-objective stages, irrelevant variables have zero reduced coefficient by (23), so none enters. If a group has several available twins, all have the same positive coefficient when that group is current and Bland selects its least member. Afterwards its other twins have reduced coefficient zero. If an earlier twin was already forbidden before the LP call, the least remaining member is selected instead. These observations prove the asserted independence from irrelevant columns and available multiplicities. $\square$

Lemma A.3 (Deletion of the objective variable). *If the column z is deleted from the restricted query, the solver returns its nonpositive certificate with zero pivots.*

Proof. All remaining original objective coefficients are zero, as are the initial slack coefficients. The all-slack objective value is zero and there is no entering variable. The stopping test precedes any pivot. $\square$

A.2 The actual block ordering and previous propagations

For the family (9), put $m = 2s - 1$. The exact matching-row order used by the model builder is

$$\begin{aligned} (Cx)_0 &= q_{0,1} + p_0 - z, \\ (Cx)_{2j-1} &= q_{j,0} - z & (1 \leq j < s), \\ (Cx)_{2j} &= q_{j,1} + p_j - z & (1 \leq j < s). \end{aligned}$$

Equivalently, the chain groups before propagation are

$$\begin{aligned} G_0 &= \{q_{0,1}, p_0\}, \\ G_{2j-1} &= \{q_{j,0}\} & (1 \leq j < s), \\ G_{2j} &= \{q_{j,1}, p_j\} & (1 \leq j < s). \end{aligned}$$

For completeness, let $n_s = 7(2s + \lceil s/4 \rceil)$ and index the original columns $v_0, \dots, v_{n_s-1}$ in increasing order. In each seven-coordinate block, the four triangle coordinates precede the three quadrilateral coordinates. With zero-based indices, the distinguished columns are

$$\begin{aligned} q_{j,0} &= v_{7(s+j)+4}, & q_{j,1} &= v_{7(s+j)+5} & (0 \leq j < s), \\ p_j &= v_{14s+7\lfloor j/4 \rfloor + (j \bmod 4)} & & & (0 \leq j < s). \end{aligned}$$

Thus $z = v_{7s+4}$, and every $q_{j,1}$ precedes its auxiliary triangle twin p_j . These are the block and column conventions of Section 6. All remaining variables are irrelevant variables of the chain lemma. The synthetic model has no triangle-anchor restrictions.

At propagation stage j , each earlier block $\ell < j$ has been forced to quadrilateral type zero. Its variable $q_{\ell,1}$ has therefore been removed, while p_ℓ remains available. Its group $G_{2\ell}$ becomes the nonempty singleton $\{p_\ell\}$. All other row groups remain nonempty. Lemma A.2 consequently gives exactly $m + 1 = 2s$ pivots for every current-node LP.

Deleting a tested irrelevant prefix coordinate removes only one member of $\mathcal{Y}$. It neither changes the objective nor empties a row group, so every such deletion query also takes exactly $2s$ pivots. Although the number of restricted original columns changes, the relative slack order used in the leaving-variable proof does not.

Deleting the mandatory coordinate $q_{j,0}$ has two different cases. For $j = 0$, this is deletion of z , so Lemma A.3 gives zero pivots. For $j > 0$, it empties G_{2j-1} and leaves all earlier groups nonempty. The first empty row is thus $h = 2j - 1$, and Lemma A.2 gives $h + 1 = 2j$ pivots.

A.3 Summing the exact costs

Proof of Theorem A.1. Theorem 6.7 establishes that both runs make $s + 1$ current-node queries and s mandatory deletion queries. The former each cost $2s$ pivots, and the latter cost $0, 2, \dots, 2(s - 1)$. The baseline additionally makes $3s^2$ positive irrelevant-prefix deletion queries, each costing $2s$ pivots.

Hence

$$P_{\text{base}}(s) = (s+1)(2s) + (3s^2)(2s) + \sum_{j=0}^{s-1} 2j = 6s^3 + 3s^2 + s,$$

$$P_{\text{screened}}(s) = (s+1)(2s) + \sum_{j=0}^{s-1} 2j = 3s^2 + s.$$

Screening these particular deletions needs only the current-node witness, so a zero-capacity persistent cache gives the same result. $\square$

The difference is exactly $6s^3$ pivots. The proof concerns the specified matrix family, column and row orders, strict entering rule, and stopping tests. It establishes neither a polynomial pivot bound for arbitrary Bland-simplex inputs nor a universal wall-clock factor. It also leaves the separate geometric realizability question unchanged.

B Claim and assumption ledger

This ledger makes the boundary of each result explicit for repository integration and later mathematical review.

Claim	Required assumptions	Evidence or proof boundary
Disjoint local moves commute	Simultaneously legal repository-style formal bipyramid replacements with disjoint consumed cells	Lemma 3.1; tracked boundary maps and cocycles; no strict-simplicial extension claimed
Constant-alphabet encoding	At most U total upward moves; exact incarnation ports	Theorem 3.3; 11^{3t+5U} , or 7^{3t} downward
Sleep search has the same count	Exact birth identities, canonical local frames, dynamic persistence of enabled independent events	Theorem 3.5; continuation states are not merged by endpoint equality
Region-sensitive search	Consumed initial footprint contained in an allowed r -cell region with at most c components	Theorem 4.5; constructive enumeration; actual footprint can have more components than its cover
Quasi-polynomial search class	$c = O(\log t)$, $r + U = O(\log^2 t)$; polynomial complete endpoint predicate	Corollary 4.6; no universal coverage of unknots supplied
Binary marking control	Prescribed relative-boundary height transport	Lemma 5.1; $B + U + O(1)$ bits; component-predicate cost remains explicit
Diagram-bound positive verdict	Authenticated exterior, replayed changes and an independently verified essential-disc component	Existing unchanged diagram verifier; every nonpositive bounded outcome is inconclusive
Exact LP certificate preservation	Fixed deterministic engine/policy; full support validation; no external time/callback cutoff for budget comparison	Theorem 6.4; node/depth/pivot completion preservation
Support-sensitive sweep	Pinned verified current witness, even with persistent cache capacity zero	Corollary 6.5; does not bound number of branch nodes
Exponential support antichain	Declared sparse algebraic cone	Proposition 6.3; no triangulation realization needed for the algebraic limitation

Claim	Required assumptions	Evidence or proof boundary
Quadratic-to-linear LP calls	Explicit seven-block model and original column order	Theorem 6.7; $3s^2 + 2s + 1$ versus $2s + 1$
Cubic-to-quadratic pivots	Same model plus the exact ordered all-slack Bland tableau and stopping rule	Theorem A.1; full induction; not a general bit-time theorem
Measured geometric speedup	Six declared inverse-bipyramid controls and three recorded pairs per size	Raw source/certificate/timing files; 22.18 paired ratio at $p = 6$; small case regression retained
Measured LP improvement	Fifteen declared cases, matched sources and explicit resource caps	Completed genuine certificates independently replay; figure-eight runs remain inconclusive
General quasi-polynomial recognition	Uniform source construction and coverage hypotheses of Proposition 9.1	Conditional consequence only; this article does not prove those hypotheses

B.1 Useful counterexamples to tempting shortcuts

The following examples delimit the mechanisms rather than merely lowering reported success rates.

1. A coherent disc can transport to a disc plus spheres. Therefore total Euler characteristic and preserved cohomology do not certify preserved surface topology.
2. A surviving tetrahedron can change row number after a disjoint move. Therefore row-number action identities are not stable enough for sleep pruning or birth-name equivalence.
3. An event can consume the output of a previous event while their consumed input sets are disjoint. Therefore historical input disjointness is not by itself a static independence relation.
4. Two equal endpoint triangulations can carry different remaining upward budgets or sleep restrictions. Therefore endpoint-query deduplication does not authorize merging their continuations.
5. A positive witness valid under one triangle anchor can violate another anchor. Therefore checking only its tested quadrilateral coordinate is insufficient for cross-query reuse.
6. A sparse unit-coefficient cone can have 2^k minimal positive supports. Therefore an unlimited witness antichain is not automatically polynomial.
7. The cancelled-braid unknot can exhaust the tested empty-region family without a disc certificate. Therefore finite-family exhaustion is not an unconditional negative knot verdict.
8. Native simplification can yield different finite triangulations of the same manifold. Therefore a fresh construction should not be required to equal a saved combinatorial fixture unless determinism has been proved.

B.2 Research status

The explicit proofs received separate internal mathematical and implementation reviews. The finite audit uses independent replay and an optional external topology oracle. This report and its code were prepared with AI assistance for ProveIt and are submitted for external mathematical and code review.

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